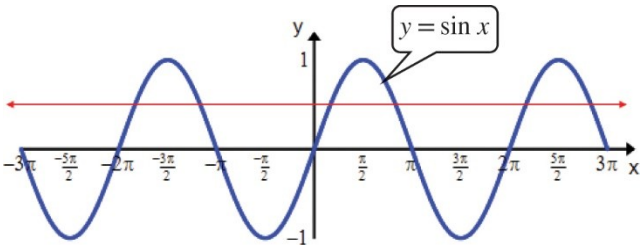
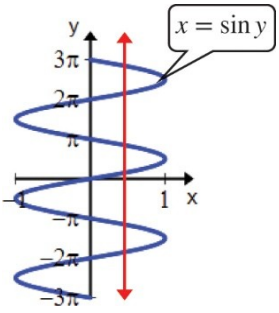
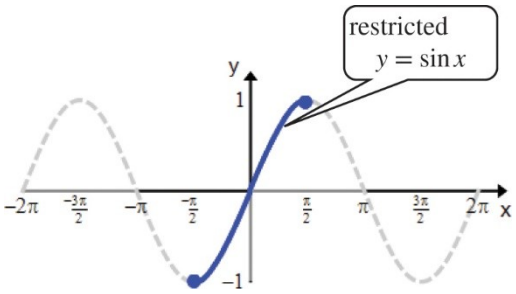
Evaluate the Inverse Sine Function





Recall that a function must be one-to-one to have an inverse *function*. For $y = \sin x$, any horizontal line taken between $-1 \le y \le 1$ intersects the graph infinitely many times. Therefore, $y = \sin x$ is not a one-to-one function.

Restrict the domain of $y = \sin x$ to the interval $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.
The graph of the restricted sine function *is* one-to-one and contains $-1 \le y \le 1$ the entire range of y values .



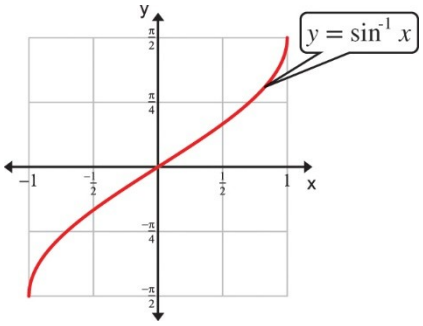
The inverse of this restricted sine function is called the inverse sine function and is denoted by $\sin^{-1}$ or $\arcsin$.

Inverse sine function

$y = \sin^{-1} x \quad \Leftrightarrow \quad \sin y = x$

$y = \arcsin x \quad \Leftrightarrow \quad \sin y = x$ where $-1 \le x \le 1$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

To evaluate $y = \sin^{-1} x$ or $y = \arcsin x$ means to find an angle y between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, inclusive, whose sine value is x.



Evaluate.

1. $\sin^{-1}\left(-\frac{1}{2}\right) = y$

$\sin y = -\frac{1}{2}$

$y = -\frac{\pi}{6}$

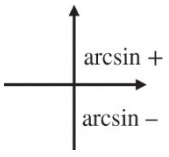
$\therefore \sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$

2. $\arcsin\frac{\sqrt{3}}{2} = y$

$\sin y = \frac{\sqrt{3}}{2}$

$y = \frac{\pi}{3}$

$\therefore \arcsin\frac{\sqrt{3}}{2} = \frac{\pi}{3}$



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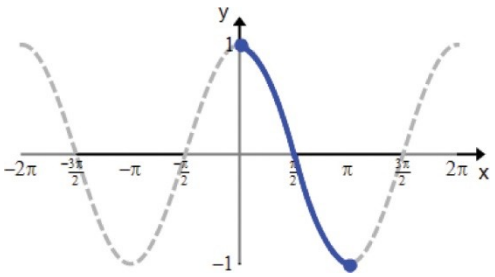
$\sin^{-1}\frac{5\pi}{4} = y$

3. $\sin y = \frac{5\pi}{4} > 1 \quad \therefore \sin^{-1}\frac{5\pi}{4}$ does not exist.

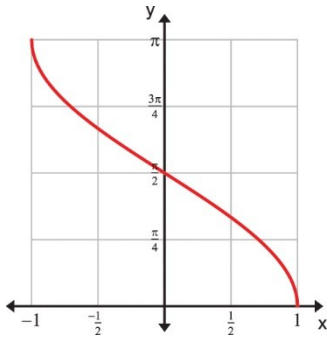
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Evaluate the Inverse Cosine and Tangent Functions

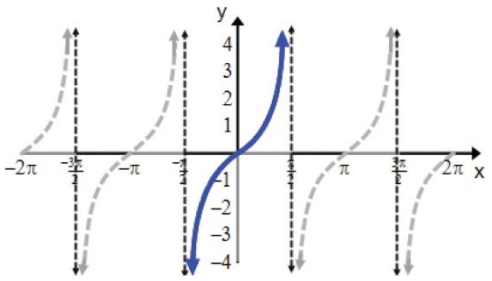
Restricted $y = \cos x$



$y = \cos^{-1} x \quad -1 \le x \le 1$ and $0 \le y \le \pi$

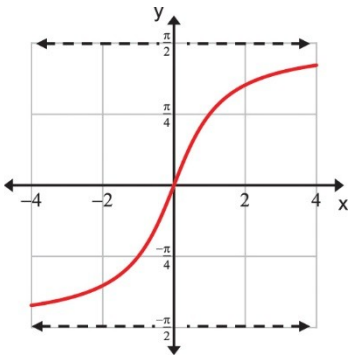


Restricted $y = \tan x$



$y = \tan^{-1} x \quad x \in \mathbb{R}$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$

Horizontal asymptotes: $y = -\frac{\pi}{2}, \; y = \frac{\pi}{2}$



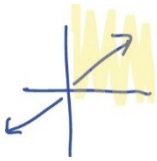
Evaluate.

4. $\arctan 1 = y$

$\tan y = 1$

$y = \frac{\pi}{4}$

$\therefore \arctan 1 = \frac{\pi}{4}$



5. $\cos^{-1}\left(-\frac{1}{2}\right) = y$

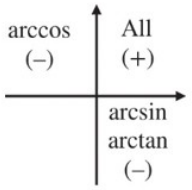
$\cos y = -\frac{1}{2}$

$y = \frac{2\pi}{3}$

$\therefore \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$



remember the signs



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6. $\arccos\frac{\sqrt{3}}{2} = y$

$\cos y = \frac{\sqrt{3}}{2}$

$y = \frac{\pi}{6}$

$\therefore \arccos\frac{\sqrt{3}}{2} = \frac{\pi}{6}$



7. $\tan^{-1} 0 = y$

$\tan y = 0$

$y = 0$

$\therefore \tan^{-1} 0 = 0$

Approximate Inverse Trigonometric Functions on a Calculator

Use a calculator to approximate the function values in radians or degrees.

8. $\arccos\frac{3}{5} \approx 53.1°$ or 0.9273 (radians)

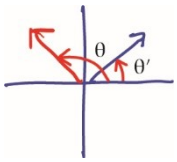
9. $\sin^{-1}(-0.457) \approx -27.2°$ or -0.4746 (radians)

10. $\arctan(-2.09) \approx 64.4°$ or -1.1245 (radians)

11. Use a calculator to approximate the degree measure (to 1 decimal place) of the angle θ where $\sin\theta = \frac{3}{8}$, $90° < \theta < 180°$.

$\theta' = \sin^{-1}\frac{3}{8}$

$\theta' \approx 22.0°$



$\theta = 180° - 22.0°$

$= 158.0°$

Compose Trigonometric Functions and Inverse Trigonometric Functions

$\sin\left(\sin^{-1} x\right) = x$ for $-1 \le x \le 1$

$\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

$\tan\left(\tan^{-1} x\right) = x$ for $x \in \mathbb{R}$

$\tan^{-1}(\tan x) = x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$\cos\left(\cos^{-1} x\right) = x$ for $-1 \le x \le 1$

$\cos^{-1}(\cos x) = x$ for $0 \le x \le \pi$

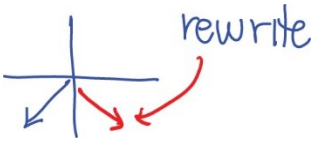
Evaluate.

12.

a. $\arcsin\left(\sin\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$

$-\frac{\pi}{2} \le -\frac{\pi}{6} \le \frac{\pi}{2}$ ✓

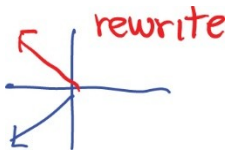
b. $\arcsin\left(\sin\frac{7\pi}{6}\right) = \arcsin\left(\sin\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$



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13.

a. $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \cos^{-1}\left(\cos\frac{5\pi}{6}\right)$



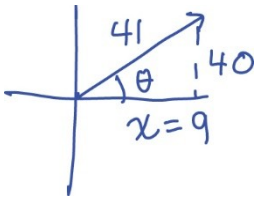
$= \frac{5\pi}{6}$

b. $\cos^{-1}\left(\cos\left(-\frac{1}{2}\right)\right) = \frac{1}{2}$

*compare to $\cos\left(\cos^{-1}\left(-\frac{1}{2}\right)\right) = -\frac{1}{2}$

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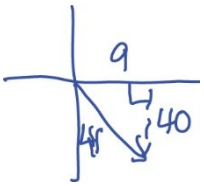
14. Find the exact value.



a. $\cos\left(\sin^{-1}\frac{40}{41}\right)$

$= \cos\theta = \frac{9}{41}$

b. $\cos\left(\sin^{-1}\left(-\frac{40}{41}\right)\right) = \cos\theta = \frac{9}{41}$



Let $\theta = \sin\left(-\frac{40}{41}\right)$, then $\sin\theta = -\frac{40}{41}$.

Let $\theta = \sin^{-1}\frac{40}{41}$, then $\sin\theta = \frac{40}{41}$.

$x^2 + 40^2 = 41^2$

$x^2 = 1681 - 1600$

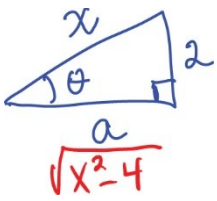
$x^2 = 81$

$x = 9$

15. Write $\tan\left(\arcsin\frac{2}{x}\right)$ for $x > 2$ as an algebraic expression.

Let $\theta = \arcsin\frac{2}{x}$

$\sin\theta = \frac{2}{x} = \frac{\text{opp}}{\text{hyp}}$



$a^2 + 2^2 = x^2$

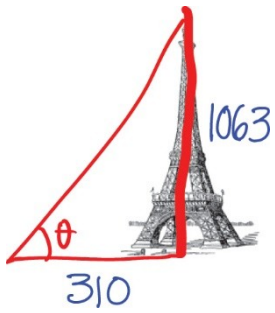
$a^2 = x^2 - 4$

$a = \sqrt{x^2 - 4}$

$\tan\left(\arcsin\frac{2}{x}\right) = \tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{2}{\sqrt{x^2-4}}$

Apply Inverse Trigonometric Functions

16. The Eiffel Tower is 1063 feet tall. A measurement is taken of the length of the shadow of the tower and is found to be 310 feet. Approximate the angle of elevation of the Sun to the nearest tenth of a degree.



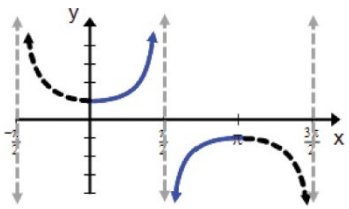
$\tan\theta = \frac{1063}{310}$

$\theta = \tan^{-1}\frac{1063}{310} \approx 73.7°$

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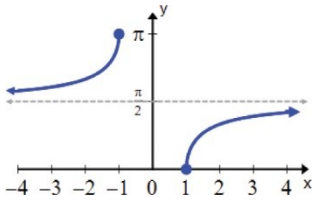
Evaluate the Inverse Secant, Cosecant, and Cotangent Functions

Restricted $y = \sec x$

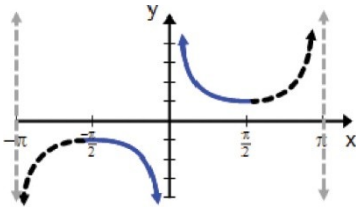


$y = \sec^{-1} x \quad \Leftrightarrow \quad \sec y = x$

where $|x| \quad 1$ and $0 \le y < \frac{\pi}{2}$ or $\frac{\pi}{2} < y \le \pi.$

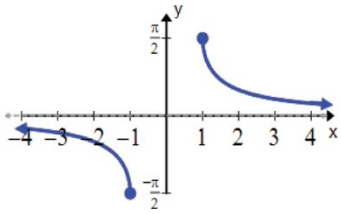


Restricted $y = \csc x$

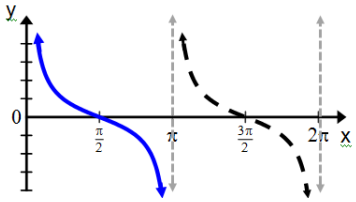


$y = \csc^{-1} x \quad \Leftrightarrow \quad \csc y = x$

where $|x| \quad 1$ and $-\frac{\pi}{2} \le y < 0$ or $0 < y \le \frac{\pi}{2}.$

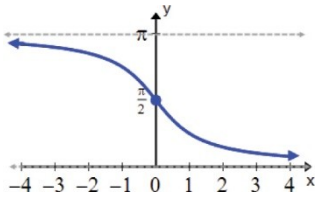


Restricted $y = \cot x$



$y = \cot^{-1} x \quad \Leftrightarrow \quad \cot y = x$

where x is any real number and $0 < y < \pi.$



Evaluate.

17. $\sec^{-1} 2 = y$

$\sec y = 2$

$\cos y = \frac{1}{2}$

$y = \frac{\pi}{3}$

$\therefore \sec^{-1} 2 = \frac{\pi}{3}$



18. $\cot^{-1}\left(-\sqrt{3}\right) = y$

$\cot y = -\sqrt{3}$

$\cot \frac{\pi}{6} = \sqrt{3}$

$\cot \frac{5\pi}{6} = -\sqrt{3}$

$y = \frac{5\pi}{6}$

$\therefore \cot^{-1}\left(-\sqrt{3}\right) = \frac{5\pi}{6}$



19. $\csc^{-1}\left(-\frac{2\sqrt{3}}{3}\right) = y$

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$$\csc y = -\frac{2\sqrt{3}}{3}$$

$$\sin y = -\frac{\sqrt{3}}{2}$$

$$y = -\frac{\pi}{3}$$

$$\therefore \csc^{-1}\left(-\frac{2\sqrt{3}}{3}\right) =$$

20. Use a calculator to approximate $\csc^{-1}\frac{8}{5}$ in radians, rounded to 4 decimal places.

$$\csc^{-1}\frac{8}{5} = y \Rightarrow \quad \csc y = \frac{8}{5}$$

no key

$$\frac{1}{\sin y} = \frac{8}{5}$$

$$\sin y = \frac{5}{8}$$

$$\sin^{-1}\frac{5}{8} \approx 0.6751$$

$$\therefore \csc^{-1}\left(\frac{8}{5}\right) \approx 0.6751$$

use

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